From the highlighted identity to equation (A)

Setting. Let $B_k \in \mathbb{R}^{n\times n}$ be symmetric, $\lambda^* \in \mathbb{R}$, and $p, p^* \in \mathbb{R}^n$. Define

$$M := B_k + \lambda^* I,$$

which is symmetric because B_k and I are.

Lemma 1.

$$-(p^*)^T M(p-p^*) + \tfrac{1}{2}p^T B_k p - \tfrac{1}{2}(p^*)^T B_k p^* \;=\; \tfrac{1}{2}(p-p^*)^T M(p-p^*) + \tfrac{\lambda^*}{2}\left(\|p^*\|^2 - \|p\|^2\right).$$

Consequently, the highlighted expression for $m_k(p) - m_k(p^*)$ equals (A).

Proof. Denote the left-hand side by L.

Step 1 (replace B_k by M in the quadratic terms). Since $B_k = M - \lambda^* I$, for any $v \in \mathbb{R}^n$

$$v^T B_k v = v^T M v - \lambda^* v^T v = v^T M v - \lambda^* \|v\|^2.$$

Applying this with $v = p$ and $v = p^*$,

$$\tfrac{1}{2}p^T B_k p - \tfrac{1}{2}(p^*)^T B_k p^* = \tfrac{1}{2}p^T M p - \tfrac{1}{2}(p^*)^T M p^* + \tfrac{\lambda^*}{2}\left(\|p^*\|^2 - \|p\|^2\right).$$

Hence

$$L = \underbrace{-(p^*)^T M(p-p^*) + \tfrac{1}{2}p^T M p - \tfrac{1}{2}(p^*)^T M p^*}_{=:Q} + \tfrac{\lambda^*}{2}\left(\|p^*\|^2 - \|p\|^2\right). \tag{1}$$

Step 2 (complete the square). Expanding and using the symmetry of M, which gives $p^T M p^* = (p^*)^T M p$ (a scalar equals its transpose),

$$\tfrac{1}{2}(p-p^*)^T M(p-p^*) = \tfrac{1}{2}p^T M p - (p^*)^T M p + \tfrac{1}{2}(p^*)^T M p^*.$$

Writing $-(p^*)^T M p = -(p^*)^T M(p-p^*) - (p^*)^T M p^*$,

$$\tfrac{1}{2}(p-p^*)^T M(p-p^*) = -(p^*)^T M(p-p^*) + \tfrac{1}{2}p^T M p - \tfrac{1}{2}(p^*)^T M p^* = Q.$$

Step 3. Substituting $Q = \tfrac{1}{2}(p-p^*)^T M(p-p^*)$ into (**??**),

$$L = \tfrac{1}{2}(p-p^*)^T (B_k + \lambda^* I)(p-p^*) + \tfrac{\lambda^*}{2}\left(\|p^*\|^2 - \|p\|^2\right),$$

which is (A). □

Remark. Symmetry of B_k is used exactly once, in Step 2, to merge the cross terms $p^T M p^*$ and $(p^*)^T M p$ into $2(p^*)^T M p$. Without symmetry the square cannot be completed in this form.